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B.Tech. Degree VII Semester Supplementary Examination May 2024

ME 19-205-0705 SUPPLY CHAIN MANAGEMENT (2019 Scheme)

Time: 3 Hours

Maximum Marks: 60

Course Outcome

On successful completion of the course, the students will be able to:

- CO1: Define a supply chain and understand its different structures and its importance to the success of a firm.
 CO2: Understand the concept of strategic fit between the SC strategy and the competitive strategy of the firm and how to achieve it.
 CO3: Identify the main drivers of supply chain (SC) performance and measure them using precise metrics.
 CO4: Assess the importance of distribution networks and the different opinions available for their design.
 CO5: Develop a methodological framework for network design and facility location and use optimisation models and techniques for facility location and capacity allocation.
 CO6: Use safety inventory to deal with demand uncertainty in the supply chain.
 CO7: Understand the importance of transportation in the SC, identify the different modes of transportation and the role played by the infrastructure and policies.

Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze, L5 – Evaluate, L6 – Create

PO – Programme Outcome

PART A(Answer *ALL* questions)

(8 × 3 = 24)

		Marks	BL	CO	PO
I.	(a) Consider the supply chain for canned tomato sauce sold by a major food processing company. What are the sources of uncertainty in this supply chain?	3	L4	4	1
	(b) Why is it important for a firm to periodically review its logistics network change over time?	3	L2	4	1
	(c) Define service level. What is its importance in inventory management?	3	L2	4	1
	(d) Explain ordering cost and carrying cost.	3	L2	4	1
	(e) Discuss the advantages of a push-base supply chain.	3	L2	4	1
	(f) Discuss the trade-off between product quality and price in traditional and inline retailing.	3	L3	6	2
	(g) Is the ERP industry fragmented? Explain why or why not.	3	L2	2	1
	(h) How is managing a multi-planned international network different from managing a domestic multi-planned network?	3	L2	2	1

PART B

(4 × 12 = 48)

II.	(a) Discuss the goal of a supply chain and explain the impact of supply chain decisions on the success of a firm.	4	L1	3	1
	(b) Describe the pull view of a supply chain.	4	L3	3	2
	(c) What are some strategic, planning and operational decisions that must be made by an apparel retailer like Fabindia?	4	L2	2	3

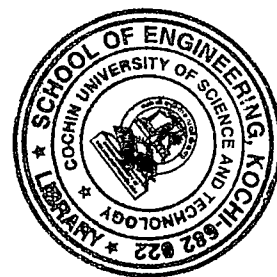
OR

III.	(a) Describe how a company achieves strategic fit between its supply chain strategy and its competitive strategy.	4	L1	4	1
	(b) Explain any two major drivers of supply chain performance.	4	L3	4	2
	(c) How can a home delivery company use pricing of its delivery services to improve its profitability?	4	L3	7	2

(P.T.O.)

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		Marks	BL	CO	PO
IV.	(a) Explain how demand uncertainty is analysed in inventory management.	3	L2	4	1
	(b) How can firms cope with huge variability in customer demand?	3	L3	4	2
	(c) A regional warehouse purchases hand tools from various suppliers and then distributes them on demand to retailers in the region. The warehouse operates 5 day a week, 52 weeks per year. The following data are estimated for a 1 inch drill bit. Average daily demand = 100 Standard deviation of daily demand = 30 Supplier lead time = 3 days Holding cost = ₹9.4 per unit per year Ordering cost = ₹35 per order Service level = 98% Design an inventory for this project.	6	L5	6	2
OR					
V.	(a) Aster Medicity orders its antibiotics every 4 weeks when a sales person from a pharmaceutical company visits it. Zombaciline, which costs ₹20 per capsule is one of its most prescribed antibiotics, with an average daily demand of 60 capsules. The standard deviation of daily demand derived from examining prescriptions filled over the past 6 months found to be 14 capsules. It takes 2 weeks for an order to arrive. The hospital would like 99% of all demands from prescriptions to be satisfied from stock. The cost to place an order is ₹1000 and the holding cost is 20% of the purchase price. The sales person has just arrived and there are currently 1000 capsules in stock. How many capsules should be ordered?	7	L5	6	2
	(b) What factors should management consider when determining a target service level?	5	L2	6	2
VI.	(a) Compare and contrast issues involved in vehicle routing in a food delivery company.	4	L3	7	2
	(b) How do transport company benefit from a technology like global positioning system (GPS)?	4	L3	7	1
	(c) What are the difficulties in implementing cross docking?	4	L3	7	2
OR					
VII.	(a) What are the main drivers of trucking rate structures?	6	L2	7	2
	(b) A firm wants to design incentive systems for its transport contractor so as to improve transport lead time reliability. To design this system they want to quantify the benefits of transport lead time reliability. Suggest a suitable approach for this quantification process.	6	L4	7	3
VIII.	(a) What are the key functional roles of IT in a supply chain?	4	L2	5	1
	(b) How can web services help a company to communicate with its suppliers and customers?	4	L3	5	2
	(c) Explain the impact of ERP implementation using financial data.	4	L2	5	2
OR					
IX.	(a) How can one implement the concept of supply chain integrations within the planning systems of a firm?	6	L2	5	2
	(b) What issues must be considered when designing supply chain relationships to improve the chances of developing cooperation and trust?	6	L3	5	3



Blooms's Taxonomy Level's

L1 – 6.7%, L2 – 41.7%, L3 – 33.3%, L4 – 5%, L5 – 13.3%.
